IE206 Complete Notes (from Provided PDFs)

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1 Coverage Map (All Provided Files)

Source PDF	Topics included in these notes
`IE206_Lec01.pdf`	Supervised setup, classification vs regression, simple/multiple linear regression, MSE/LSE, correlation, R^2, SST/SSM/SSE, polynomial fit
`IE206_Logistics_2.pdf`	Model validation perspective, regression diagnostics, outliers/influence, robust losses (L2/L1/Huber/Bisquare), Ridge/LASSO, Bayesian view of regularization, logistic regression and sigmoid/logit
`IE_206_ModelEvaluation_3.pdf`	Hypothesis classes, train/test split, confusion matrix metrics, precision/recall/F1, underfit/overfit, holdout, k-fold CV, model selection curve
`IE_206_SGD.pdf`	GD/SGD basics, subgradients, hinge subgradient, step-size schedules, strongly convex case, projected SGD, convergence guarantee in convex-Lipschitz bounded setting
`IE_206_Perceptron_4.pdf`	Halfspaces, perceptron update rule, convergence theorem and mistake bound, feature scaling/standardization, feature engineering and polynomial basis
`IE_206_SVM.pdf`	Linear separability, margin geometry, hard/soft SVM, slack variables, hinge-loss regularization, support vectors, duality, dual optimization
`IE_206_KNN.pdf`	KNN algorithm, distance metrics, curse of dimensionality, Bayes classifier/risk/optimality, Naive Bayes, discriminative vs generative methods
`IE_206_MultiClass.pdf`	One-vs-All, All-Pairs, reduction pitfalls, linear multiclass predictors, multiclass SVM, SGD update, ranking intro and losses
`IE_206_RandoMForest.pdf`	Decision trees, ID3, entropy/gini gains, threshold splits for real-valued features, random forest construction, boosting, decision stumps, ERM for stumps, AdaBoost and training error bound
`IE206_PracticeQs_1.pdf`	Cost-sensitive model choice, imbalanced metrics tradeoff, LOOCV in linear regression, hyperparameter selection, Sherman-Morrison identity, fast LOOCV formula via hat matrix, convexity/nondifferentiability of clipped regression loss
`IE206_PracticeQs_2.pdf`	Bayes vs 1-NN error bound, scaling effect in KNN, hard-margin SVM worked geometry, soft-margin extremes ($C = 0, \infty$), KKT interpretation ($\alpha_i = C$ case)

2 Linear Regression Notes (`IE206_Lec01.pdf`)

Learning setup

Given i.i.d. samples

$$S = \{(x^{(i)}, y^{(i)})\}_{i=1}^{m},$$

classification has finite/discrete label set $\mathcal{Y}$, while regression has continuous (or non-countable) labels.

Simple linear regression

Model:

$$y = \beta_0 + \beta_1 x.$$

Probabilistic form with noise:

$$y^{(i)} = \beta_0 + \beta_1 x^{(i)} + \varepsilon_i, \qquad \mathbb{E}[y^{(i)} \mid x^{(i)}] = \beta_0 + \beta_1 x^{(i)}.$$

Goodness of fit and least squares

Residual $r_i = y^{(i)} - (\beta_0 + \beta_1 x^{(i)})$. Common losses:

- absolute error: $|r_i|$
- squared error: r_i^2

MSE/LSE objective:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg\min_{\beta_0,\beta_1} \sum_{i=1}^{m} r_i^2.$$

Closed form:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Correlation and interpretation

Sample correlation coefficient:

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\sqrt{\sum_i (y_i - \bar{y})^2}}.$$

Interpretation: sign gives direction, magnitude gives linear association strength.

Multiple linear regression

$$y^{(i)} = x^{(i)\top}\beta + \varepsilon_i, \quad \hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{m} (y^{(i)} - x^{(i)\top}\beta)^2.$$

Matrix form (standard convention with $X \in \mathbb{R}^{m\times d}$):

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

(when $X^\top X$ is invertible).

Coefficient of determination and decomposition

$$\mathrm{SST} = \sum_i (y_i - \bar{y})^2, \quad \mathrm{SSE} = \sum_i (y_i - \hat{y}_i)^2, \quad \mathrm{SSM} = \sum_i (\hat{y}_i - \bar{y})^2.$$

Identity:

$$\mathrm{SST} = \mathrm{SSM} + \mathrm{SSE}, \qquad R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}} = \frac{\mathrm{SSM}}{\mathrm{SST}}.$$

Polynomial fit (feature expansion)

Linear in parameters can represent nonlinear curves by expanding features:

$$\phi(x) = (1, x, x^2, \ldots, x^p), \quad y \approx w^\top \phi(x).$$

Important caution

High training fit does not imply good extrapolation/generalization; always validate on unseen data.

3 Logistic + Robust Regression Notes (`IE206_Logistics_2.pdf`)

Regression diagnostics and outliers

Key checks:

- residual behavior (random vs patterned)
- outliers (large residual) and influential points (large effect on fitted model)
- feature redundancy/collinearity

Robustness through loss design

Let $r_i = y_i - x_i^\top \beta$.

- L2 (squared): $\rho(r) = r^2$ (sensitive to large outliers)
- LAD/L1: $\rho(r) = |r|$ (more robust; nondifferentiable at 0)
- Huber:

$$\rho_k(r) = \begin{cases} \frac{1}{2} r^2, & |r| \le k, \\ k\left(|r| - \frac{k}{2}\right), & |r| > k, \end{cases}$$

  quadratic near 0 and linear in tails
- Bisquare/Tukey: strong downweighting of very large residuals

Distribution view of loss choice

- Gaussian noise $\Rightarrow$ squared loss (MLE)
- Laplace noise $\Rightarrow$ absolute loss (MLE)

Ridge and LASSO

Ridge:

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta} \sum_{i=1}^{m} (y_i - x_i^\top \beta)^2 + \lambda \sum_{j=1}^{d} \beta_j^2$$

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y.$$

Effects: shrinkage, handles collinearity, usually not sparse.

LASSO:

$$\hat{\beta}_{\text{lasso}} = \arg\min_{\beta} \sum_{i=1}^{m} (y_i - x_i^\top \beta)^2 + \lambda \sum_{j=1}^{d} |\beta_j|.$$

Effects: shrinkage + feature selection (sparse coefficients).

Bayesian connection

MAP estimation with priors:

- Gaussian prior on $\beta \Rightarrow$ Ridge penalty
- Laplace prior on $\beta \Rightarrow$ LASSO penalty

Logistic regression

Binary classification with score $\eta = x^\top \beta$:

$$\mathbb{P}(Y = 1 \mid X = x) = \sigma(\eta) = \frac{1}{1 + e^{-\eta}}.$$

Logit transform:

$$\log \frac{p(x)}{1 - p(x)} = x^\top \beta.$$

Likelihood for data $\{(x_i, y_i)\}$, $y_i \in \{0, 1\}$:

$$L(\beta) = \prod_{i=1}^{m} p_i^{y_i} (1 - p_i)^{1 - y_i}, \quad p_i = \sigma(x_i^\top \beta).$$

Equivalent minimization is negative log-likelihood (logistic loss), solved iteratively (no closed-form like OLS).

4 Model Evaluation Notes (`IE_206_ModelEvaluation_3.pdf`)

Hypothesis class

A hypothesis is a mapping $h : \mathcal{X} \to \mathcal{Y}$. Examples: linear, quadratic, hyperplane, polynomial classes. Learning chooses a good h from $\mathcal{H}$.

Train-test split

Typical split: train 70–80%, test 20–30%. Test data estimates out-of-sample performance.

Confusion matrix metrics

With TP, TN, FP, FN:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN},$$

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN},$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Use precision when FP cost is high; recall when FN cost is high; F1 for class imbalance/tradeoff.

Underfitting vs overfitting

- Underfit: high train and test error (high bias)
- Proper complexity: low test error
- Overfit: low train but high test error (high variance)

Validation strategies

- Holdout: simple, but can waste data and have high variance estimate
- k-fold CV: each sample validated once, more stable estimate
- Practical default often $k = 5$ or 10 depending on data size/compute

Model selection should be based on validation/CV performance, not training error.

5 SGD Notes (`IE_206_SGD.pdf`)

Gradient descent

For differentiable $f : \mathbb{R}^d \to \mathbb{R}$:

$$w_{t+1} = w_t - \eta \nabla f(w_t).$$

η controls step size/stability.

Subgradient for non-differentiable convex functions

A vector g is subgradient at w if

$$f(u) \geq f(w) + \langle g, u - w \rangle \quad \forall u.$$

Used when gradient does not exist (e.g., hinge, absolute value).

Hinge loss and Lipschitz idea

For labeled example (x, y):

$$\ell_{\text{hinge}}(w; (x, y)) = \max(0, 1 - y\langle w, x\rangle).$$

Subgradient:

$$\partial \ell = \begin{cases} \{0\}, & y\langle w, x\rangle > 1, \\ \{-yx\}, & y\langle w, x\rangle < 1, \\ \operatorname{conv}\{0, -yx\}, & y\langle w, x\rangle = 1. \end{cases}$$

Stochastic gradient descent

Use unbiased stochastic (sub)gradient v_t:

$$\mathbb{E}[v_t \mid w_t] \in \partial f(w_t), \qquad w_{t+1} = w_t - \eta_t v_t.$$

Common output is averaged iterate $\bar{w} = \frac{1}{T}\sum_{t=1}^{T} w_t$.

Variants

- Decreasing step size, e.g. $\eta_t \propto 1/\sqrt{t}$
- Strongly convex case often uses $\eta_t \propto 1/(\lambda t)$
- Projected SGD enforces constraints by projection:

$$w_{t+1} = \Pi_{\mathcal{H}}(w_t - \eta_t v_t).$$

Convergence message from lecture

For convex-Lipschitz bounded setup, SGD with suitable T and η reaches expected suboptimality $\leq \varepsilon$ with sample complexity matching regularized ERM order.

6 Perceptron + Feature Engineering Notes (`IE_206_Perceptron_4.pdf`)

Halfspaces and affine score

$$h_{w,b}(x) = \langle w, x\rangle + b, \qquad \hat{y} = \operatorname{sign}(h_{w,b}(x)) \in \{-1, +1\}.$$

Perceptron update rule

Given a mistake on (x_i, y_i) where $y_i\langle w_t, x_i\rangle \leq 0$:

$$w_{t+1} = w_t + y_i x_i.$$

Repeat until no mistakes (if separable).

Convergence guarantee

For separable data with

$$R = \max_i \|x_i\|, \quad B = \min\{\|w\| : \ y_i\langle w, x_i\rangle \geq 1 \ \forall i\},$$

lecture bound:

$$T \leq (RB)^2.$$

Practical implication: feature scaling strongly affects convergence speed.

Normalization and centralization

Standardization (z-score form):

$$\phi(x) = \frac{x - \bar{x}}{\sigma}.$$

Benefits: balanced feature influence, faster optimization, better numerical behavior.

Feature engineering and polynomial basis

Map original input into richer features to make classes linearly separable:

$$\phi(x_1, x_2) = (1, x_1, x_2, x_1^2, x_1x_2, x_2^2, \dots).$$

XOR-style patterns become separable in transformed space.

7 SVM Notes (`IE_206_SVM.pdf`)

Margin geometry

Distance from point x to hyperplane (w, b):

$$\mathrm{dist}(x, \{z : w^\top z + b = 0\}) = \frac{|w^\top x + b|}{\|w\|}.$$

For canonical SVM constraints $y_i(w^\top x_i + b) \geq 1$, margin width is

$$\frac{2}{\|w\|}.$$

Hard-margin SVM

$$\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{s.t.} \quad y_i(w^\top x_i + b) \geq 1 \ \forall i.$$

Needs linear separability.

Soft-margin SVM

Introduce slack $\xi_i \geq 0$:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_i \xi_i \quad \text{s.t.} \quad y_i(w^\top x_i + b) \geq 1 - \xi_i.$$

Equivalent regularized hinge-loss form:

$$\min_{w,b} \lambda\|w\|^2 + \frac{1}{m}\sum_i \max\{0, 1 - y_i(w^\top x_i + b)\}.$$

Support vectors

Points with active margin constraints determine the separator; in dual view, these are points with nonzero dual coefficients.

Duality and dual optimization

For hard margin, dual (schematic):

$$\max_{\alpha \geq 0} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle.$$

Key implications:

- solution is linear combination of training points
- optimization depends only on inner products (kernel entry point)
- weak duality always; strong duality for convex feasible settings

8 KNN + Bayes Notes (`IE_206_KNN.pdf`)

K-Nearest Neighbors

Workflow:

1. choose k and a distance metric
2. compute distance from query to all train points
3. pick k nearest points
4. classify by majority vote (or average for regression)

   Common distances for $x, y \in \mathbb{R}^n$:

$$\|x - y\|_2, \quad \|x - y\|_1, \quad \|x - y\|_p.$$

   Limitations from lecture:

- curse of dimensionality
- prediction-time cost grows with dataset size
- sensitivity to noisy data and scaling

Bayes classifier

Bayes theorem:

$$\mathbb{P}(Y \mid X) = \frac{\mathbb{P}(X \mid Y)\mathbb{P}(Y)}{\mathbb{P}(X)}.$$

Optimal classifier:

$$h^*(x) = \arg\max_y \mathbb{P}(Y = y \mid X = x).$$

Binary case:

$$h^*(x) = \mathbf{1}\{\eta(x) > 0.5\}, \quad \eta(x) = \mathbb{P}(Y = 1 \mid X = x).$$

Bayes risk (minimum possible classification error):

$$R^* = \mathbb{E}_X \left[1 - \max_k \mathbb{P}(Y = k \mid X) \right].$$

Naive Bayes

Conditional independence assumption:

$$\mathbb{P}(Y \mid X_1, \ldots, X_n) \propto \mathbb{P}(Y) \prod_{j=1}^{n} \mathbb{P}(X_j \mid Y).$$

Fast, scalable, often strong baseline (especially text tasks).

Discriminative vs generative

- Discriminative: model $\mathbb{P}(Y \mid X)$ or decision boundary directly (e.g., SVM, logistic)
- Generative: model joint/conditional data generation (e.g., Naive Bayes)

9 Multiclass Notes (`IE_206_MultiClass.pdf`)

Problem

Predict one of k classes for each input.

Reduction methods

One-vs-All (OvA): train k binary models h_i; predict

$$\hat{y} = \arg\max_i h_i(x).$$

All-Pairs (OvO): train one binary model per class pair (i, j); aggregate pairwise scores/votes.

Pitfalls of reductions

For imbalanced/low-mass classes, OvA can output trivial all-negative model for a class, causing severe multiclass failure even when a global multiclass linear separator exists.

Linear multiclass predictor

Unified score model:

$$\hat{y} = \arg\max_{y \in \{1,\ldots,k\}} \langle w, \Psi(x, y) \rangle.$$

Advantages: one jointly trained model; supports cost-sensitive losses $\Delta(y', y)$.

Multiclass SVM + SGD

Structured hinge objective (lecture form):

$$\min_w \lambda \|w\|^2 + \frac{1}{m} \sum_{i=1}^{m} \max_{y'} \Big[\Delta(y', y_i) + \langle w, \Psi(x_i, y') - \Psi(x_i, y_i) \rangle \Big].$$

Typical SGD step uses worst offending class y^*:

$$w_{t+1} = w_t - \eta_t \big(\Psi(x, y^*) - \Psi(x, y) \big)$$

(with regularization term as applicable).

Ranking intro

Output is an ordering, not a single class. Mentioned losses: Kendall-tau pairwise disagreement and NDCG (top-heavy relevance quality).

10 Decision Trees, Random Forest, Boosting Notes (`IE_206_RandoMForest.pdf`)

Decision trees

Hierarchical split model with nodes, branches, leaves. Greedy top-down building is used because globally optimal tree search is computationally hard.

ID3-style construction

Recursive rules:

- stop if pure node
- stop if no features left; output majority label
- otherwise split using feature with best gain and recurse

Gain measures

For binary class proportion a:

$$\text{Entropy } C(a) = -a \log a - (1-a)\log(1-a),$$

$$\text{Gini } C(a) = 2a(1-a).$$

Information gain = impurity before split minus weighted impurity after split.

Real-valued features via thresholds

Convert to binary tests $\mathbf{1}[x_j < \theta]$ with candidate thresholds from sorted feature values. Efficient gain search can be implemented in about $O(dm \log m)$ (or better with incremental updates after sorting).

Random forest

Core idea: reduce variance by averaging many decorrelated trees.

- bootstrap sample of training data per tree
- random subset of features at each split
- majority vote aggregation

Small feature subset size per split helps decorrelate trees.

Boosting and decision stumps

Decision stump: one-split tree (weak learner). Boosting trains weak learners sequentially with reweighting to focus on hard examples.

ERM for decision stumps

Given weighted sample distribution D, choose feature-threshold pair minimizing weighted classification error. Sorting by each feature and sweeping thresholds yields efficient implementation.

AdaBoost

At round t:

- train weak learner h_t on distribution D_t
- weighted error ε_t
- learner weight $\alpha_t = \frac{1}{2} \log \frac{1-\varepsilon_t}{\varepsilon_t}$
- upweight misclassified examples for next round

Final model:

$$H(x) = \operatorname{sign}\left(\sum_{t=1}^{T} \alpha_t h_t(x)\right).$$

Training error bound (lecture theorem style): decays exponentially with rounds when each weak learner beats random guessing ($\varepsilon_t < 1/2$).

11 Practice Sheet Notes (`IE206_PracticeQs_1.pdf`, `IE206_PracticeQs_2.pdf`)

PracticeQs_1 themes

- **Cost-sensitive decision making**: choose model by total expected penalty, not by accuracy alone.
- **Imbalanced fraud setting**: compare accuracy vs precision/recall/F1 and pick metric matching business objective.
- **LOOCV for linear regression**: compute leave-one-out models and compare fit quality (R^2/MSE).
- **Hyperparameter tuning**: choose setting with best validation performance (generalization), avoid overfitting settings with low train but poor validation error.
- **Sherman–Morrison identity**: rank-one inverse update useful for efficient LOOCV derivation.
- **Fast LOOCV formula via hat matrix**:

  $$\hat{y}_i^{(-i)} = \frac{\hat{y}_i - h_{ii} y_i}{1 - h_{ii}}, \qquad e_i^{(-i)} = \frac{e_i}{1 - h_{ii}},$$

  so

  $$\mathrm{CV}(n) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{y_i - \hat{y}_i}{1 - h_{ii}}\right)^2.$$

- **Clipped/rectified regression loss**:

  $$\ell(w; x, y) = \left| y - \max(0, w^\top x) \right|.$$

  Convex in w as composition of convex functions with affine map plus absolute value; non-differentiable at hinge boundaries where $w^\top x = 0$ and/or where the outer absolute value argument is zero.

PracticeQs_2 themes

- **Bayes vs 1-NN asymptotics**: pointwise error bound

$$\varepsilon_{NN}(x) \leq 2\varepsilon_{Bayes}(x).$$

- **Feature scaling in KNN**: nearest neighbor can flip after normalization due to scale dominance in raw Euclidean distance.
- **Hard-margin SVM geometry**: separability proof, optimal separator, support vector identification, effect of removing non-support vectors.
- **Soft-margin extremes**: behavior for $C = 0$ (regularization dominates) and $C \to \infty$ (hard-margin-like behavior if separable).
- **KKT interpretation in soft SVM**: from

$$\alpha_i(1 - y_i f_i - \xi_i) = 0, \quad \beta_i \xi_i = 0, \quad \alpha_i + \beta_i = C,$$

  if $\alpha_i = C$ then $\beta_i = 0$; point is inside margin or misclassified (cannot be strictly outside margin with inactive slack and $\alpha_i = C$ simultaneously).

12 Quick Exam-Revision Checklist

1. Derive OLS and ridge closed forms, and know when matrix inverse fails.
2. Compute and interpret R^2, SST/SSM/SSE, and correlation.
3. Know robust losses and when to prefer L1/Huber over L2.
4. Use precision/recall/F1 correctly for imbalanced data.
5. Implement CV logic and LOOCV hat-matrix shortcut.
6. Remember perceptron mistake update and convergence condition.
7. Derive hard/soft SVM primal forms, hinge loss form, and dual interpretation.
8. Explain KNN distance scaling and curse of dimensionality.
9. Distinguish Bayes optimal, Naive Bayes, discriminative vs generative.
10. Compare OvA/OvO vs joint multiclass methods.
11. Explain random forest variance reduction mechanism.
12. State AdaBoost update logic and why training error can drop exponentially.
13. Explain SGD vs GD and when projection/subgradient is required.